Cross-Domain Review Sentiment Classification

Initial Proposal and First Draft | Option 2: Review Sentiment Analysis

Course / section: ARTI 404 / 7MA2 **Group:** 4 **Instructor:** Dr. Mohammed Mosuily

1. Team and Responsibilities

Role	Name	Student ID	Technical responsibilities
Team leader	Ali Jafar Altaweel	2240006456	MiniLM features, classifier tuning and Try it integration; coordinates submission.
Member	Mohammed Moayed Alqurini	2240007325	TF-IDF pipeline, development tuning and reproducibility checks.
Member	Danyal Salman Alhashem	2240002713	Data parsing, duplicate audit, stratified splits and data documentation.

Shared by all: evaluation, error analysis, report and presentation. Each member supplies two original labelled examples and keeps dated evidence of their contribution.

2. Purpose, Input and Output

We build an English review classifier for users who need a quick summary of customer opinions. **Input:** a short, unfamiliar review. **Output:** Positive or Negative; for example, “The food was delicious.” should return Positive. A Jupyter notebook provides a *Try it* cell that shows the prediction of both methods without retraining and gives a helpful message for empty input.

3. Challenge and Experimental Question

Challenge: transfer from product and movie reviews to restaurant reviews, a domain whose vocabulary (food, service) differs from the training domains (battery, acting, story).

Question: *Which method performs better on restaurant reviews when trained on product and movie reviews?*

Yelp performance is the main comparison; an independent in-domain test provides supporting analysis.

4. Dataset and Splits

UCI Sentiment Labelled Sentences [1]: 3,000 English sentences, nominally 1,000 each from Amazon, IMDb and Yelp (500 Positive / 500 Negative per source). https://archive.ics.uci.edu/dataset/331/sentiment+labelled+sentences

Set	Source	Target size*	Use
Training	Amazon + IMDb	1,400	Fit both methods
Development	Amazon + IMDb	300	Select settings
In-domain test	Amazon + IMDb	300	Supporting comparison
Cross-domain test	Yelp	1,000	Main comparison

* Before cleaning. Duplicate and conflicting examples are audited across all sources, including Yelp, before splitting. Amazon/IMDb splits are stratified by source and label (seed 42); actual post-cleaning counts will be reported. Neither test set is used for tuning.

5. Methods and Evaluation

- **Method 1 (traditional):** TF-IDF + Logistic Regression; TF-IDF is fitted on training text only.
- **Method 2 (pretrained):** frozen all-MiniLM-L6-v2 sentence embeddings [3] + Logistic Regression; no fine-tuning.
- **Settings:** $C \in \{0.1, 1, 10\}$ for both classifiers and TF-IDF n-grams $\in \{(1,1), (1,2)\}$, selected by development macro F1.
- **Evaluation:** accuracy and macro F1 for both methods on identical examples in each test set; at least three errors analysed (or all errors plus difficult correct cases if fewer).
- **Limitations:** in-domain vs. Yelp score changes are descriptive and do not establish semantic understanding; small data, differing test difficulty and neutral/mixed reviews limit conclusions. No accuracy outcome is assumed.

6. Related Work

Kotzias et al. [1] inferred sentence-level labels from review-level supervision; their labelled sentences motivate our dataset choice. Reimers and Gurevych [2] introduced SBERT sentence representations, motivating the pretrained representation route. Our investigation compares fixed MiniLM features [3] with TF-IDF under a held-out restaurant domain, rather than reproducing either study. Data, code, models and AI assistance will be credited; implementation begins after instructor approval.

References

[1] D. Kotzias, M. Denil, N. de Freitas and P. Smyth, “From Group to Individual Labels Using Deep Features,” in *Proc. ACM SIGKDD*, 2015, pp. 597–606, doi: 10.1145/2783258.2783380.

[2] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.

[3] Sentence Transformers, “all-MiniLM-L6-v2,” model card. https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2